

What is the model actually looking at?

Making the self-supervised 4D-STEM “grain map” explainable in physical terms

A non-technical guide to the interpretability tests · IMC & NaPHI · Daniel Khaykelson

1. The setup, and the problem

Our model (a self-supervised network called DINO) looks at the electron diffraction pattern recorded at every probe position in a 4D-STEM scan and sorts each position into one of ~ 14 groups—producing a “grain map” *with no labels and no chemistry input*. It works, but it is a black box: we do not know *which physical property* defines the groups (crystallinity? thickness? crystal orientation? something new?). The four tests below answer that, each from a different angle, so that together they triangulate a trustworthy physical explanation.

2. The four tests (plain idea + what it measures)

1. Probing — what the model pays attention to

Plain idea: if we can “read off” a known property (say crystallinity) from the model’s internal notes for each point, then the model is keeping track of it.

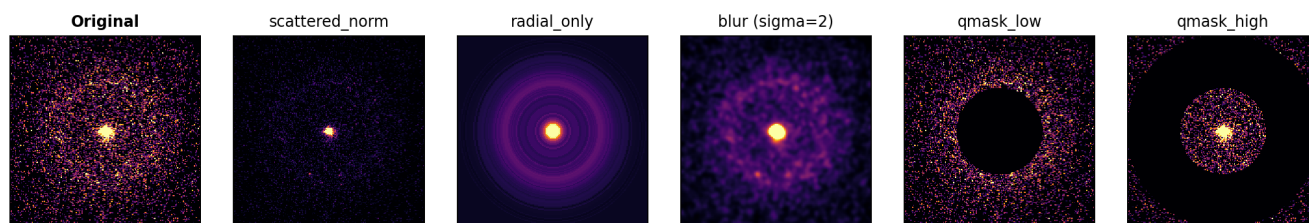
What we measure: we fit a simple linear predictor from the model’s 128-number fingerprint to each measured property; the score R^2 (0–1) says how strongly that property is encoded.

2. Ablation — what the model depends on

Plain idea: like covering part of a photo to see whether you can still recognise it.

What we measure: we blank out one ingredient of the diffraction pattern (overall brightness, fine angular detail, or inner vs. outer rings), re-run the model, and compare the new map to the original (agreement = ARI). If the map survives the edit, that ingredient was not needed; if it falls apart, the model depended on it.

The same diffraction pattern, before (*Original*) and after each input modification



The same diffraction pattern, before (*Original*) and after each modification. *scattered_norm* rescales the overall brightness away; *radial_only* keeps the rings but erases angular detail; *blur* washes out sharp spots; *qmask_low/high* blank the inner / outer rings. Each edited version is re-clustered and compared to the original map.

3. Classical check — is the map actually new?

Plain idea: does the AI just redraw what a standard, decades-old analysis already shows, or something genuinely new?

What we measure: we cluster classical 4D-STEM features (a virtual dark-field image, the azimuthally-averaged profile, and PCA/NMF decompositions of the patterns) into the same number of groups, and measure how much they agree with the AI map (ARI/AMI).

4. Orientation check — is it just sorting by crystal direction?

Plain idea: a tilted crystal diffracts differently; maybe the model is only sorting by grain *orientation*.

What we measure: where a crystal structure (CIF) is available, we index the orientation at each point (ACOM) and measure agreement with the AI map. We also use Grad-CAM to see *which part* of the pattern the model looks at.

How to read the two numbers. R^2 — how predictable a property is from the model (0 = ignored $\rightarrow$ 1 = fully encoded). **ARI / AMI** — how much two maps agree, corrected for chance (0 = no better than random $\rightarrow$ 1 = identical maps).

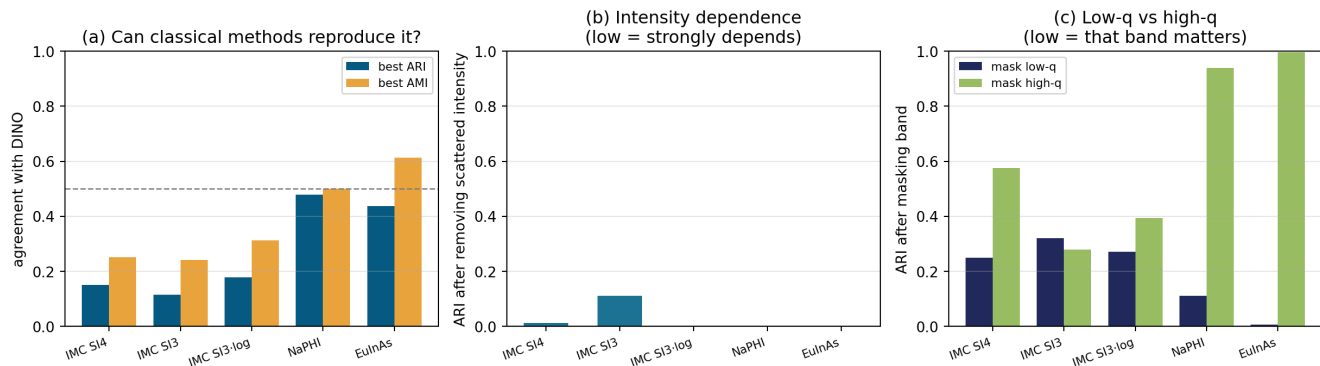
3. What we found

- The groups are driven by the **overall scattered (diffracted) intensity** together with the **inner, low-angle (low- q) 2-D diffraction structure** — the arrangement of the inner rings/spots ($q \lesssim 2.1 \text{ nm}^{-1}$, i.e. $d \gtrsim 4.7 \text{ Å}$).
- For the **molecular** films (IMC, NaPHI) it is **not** based on crystal orientation (zone-axis agreement $\text{AMI} \leq 0.06$); the model is in-plane-rotation-invariant by design.
- A 1-D radial profile alone is **not enough** — averaging away the angles collapses the map; the full 2-D pattern is needed.
- **Uniqueness is material-dependent.** For the molecular IMC films no classical method reproduces the map (best $\text{ARI} \approx 0.11\text{--}0.18$): the AI map is *distinctive*. For NaPHI and the layered EuInAs, classical methods get about halfway ($\text{ARI} \approx 0.44\text{--}0.52$, AMI up to 0.6): there the AI stays closer to a classical decomposition (see figure).
- **EuInAs is special:** being well-crystallised, its DINO classes also *partially track crystal orientation* (3-phase ACOM zone-axis $\text{AMI} \approx 0.30$ — much higher than the molecular films), while phase identity does not ($\text{AMI} \approx 0.08$). Orientation is a genuine *secondary* axis here.
- Logarithmic intensity stretching (an image-prep choice) keeps the same physical basis but *sharpens* it — more, cleaner groups.

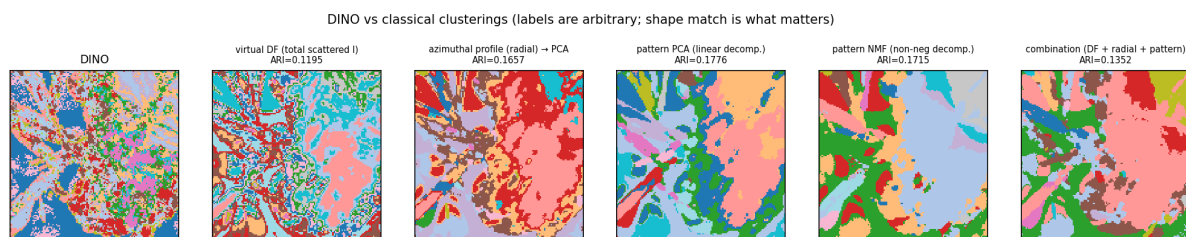
4. What it means (and one caveat)

The model groups regions by *how much, and what, they diffract at low angles* — crystallinity, molecular packing, and sample thickness. That is a physically sensible axis, not an arbitrary black box. For the molecular IMC films it is genuinely distinctive (a standard virtual-detector or PCA map does not reproduce it); for NaPHI and the layered EuInAs it stays closer to classical analysis, and EuInAs additionally tracks crystal orientation, which the molecular films do not.

Caveat: “scattered intensity” blends crystallinity with sample thickness — both raise the total diffracted signal. An independent thickness map (e.g. a t/λ estimate) would be needed to separate the two. Orientation/phase analysis is folded in automatically wherever a CIF is available; the conclusions above hold with or without it.



Cross-sample summary (IMC SI3/SI4, NaPHI, EuInAs). **(a)** how well classical methods reproduce the AI map (rises from “distinctive” IMC to “substantially captured” NaPHI/EuInAs); **(b)** intensity dependence (low bar = strongly depends); **(c)** low- q vs high- q (low bar = that band matters — NaPHI and EuInAs depend almost entirely on low- q).



IMC: the AI map (far left) vs. classical clusterings. Colours are arbitrary — what matters is the spatial shape. Classical methods catch the coarse regions but miss the fine grain structure the AI resolves.